

Two exact planar few-distance constructions

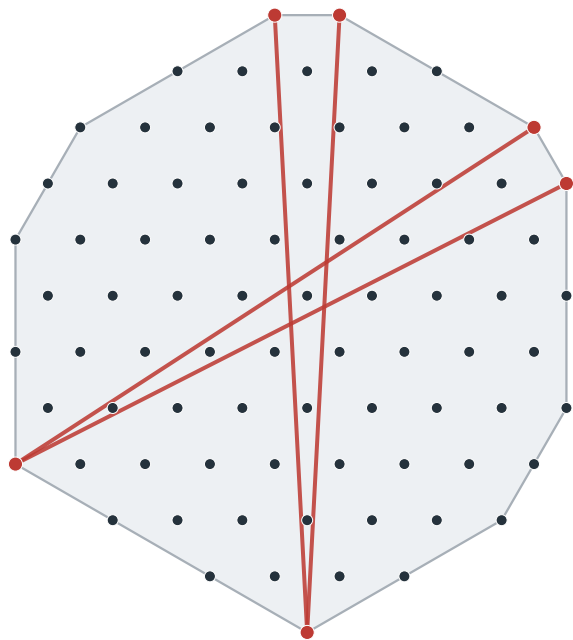
Improved construction bounds relative to Ahmed-Snevily (2013)

(a) 81 points · 31 distances

(b) 111 points · 41 distances

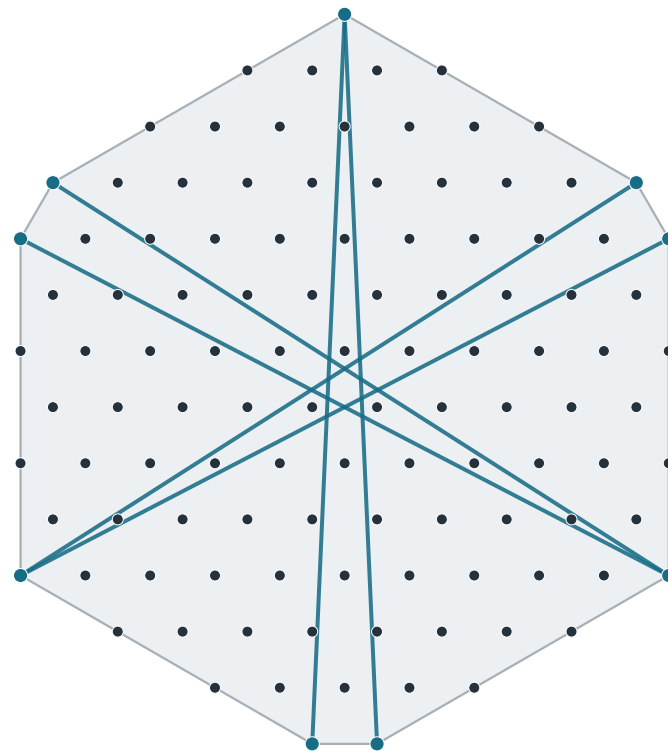
$G(31) \geq 81$ · published bound: 80

$G(41) \geq 111$ · published bound: 109



1 unit
|-----|

$d^2 = 91$: 4 highlighted pairs



$d^2 = 127$: 6 highlighted pairs

Identical unit spacing in both panels; coordinates translated for display. Faint regions are the filled convex hulls.